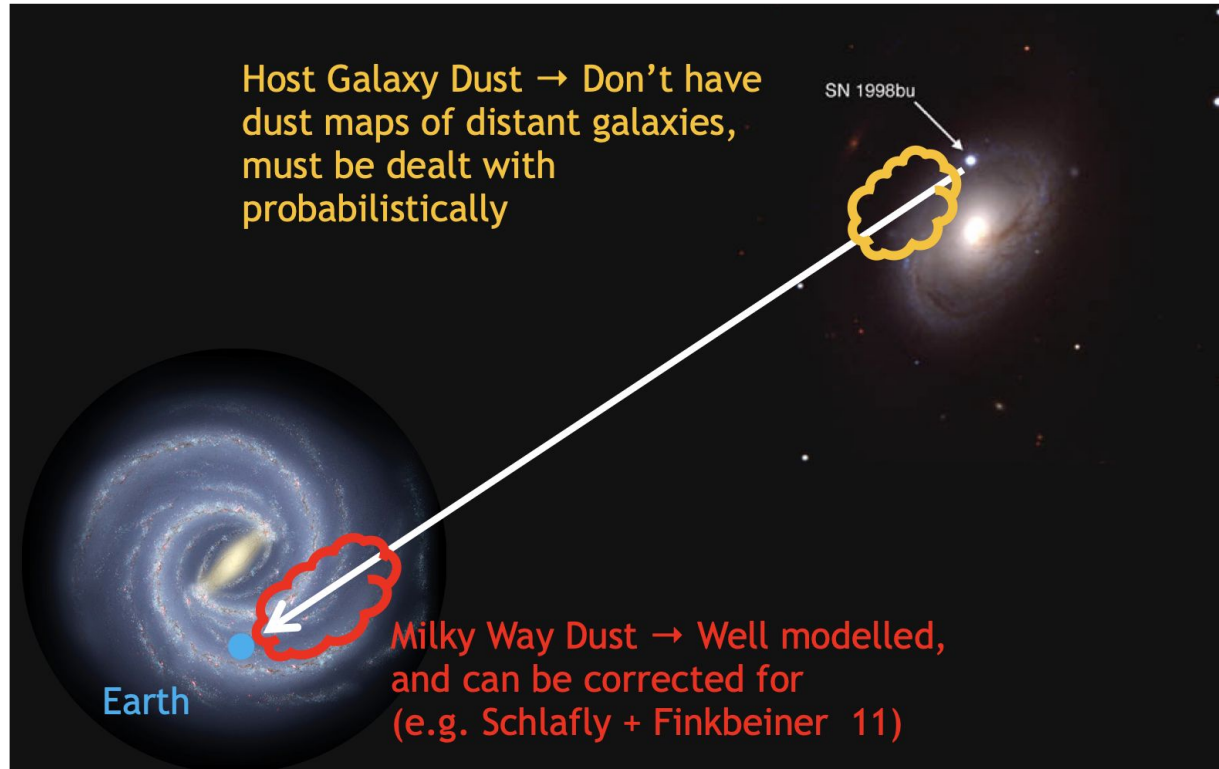


Analysing the SN Ia Mass Step using Hierarchical BayeSN Inference

Matt Grayling, Stephen Thorp, Kaisey Mandel



Motivation



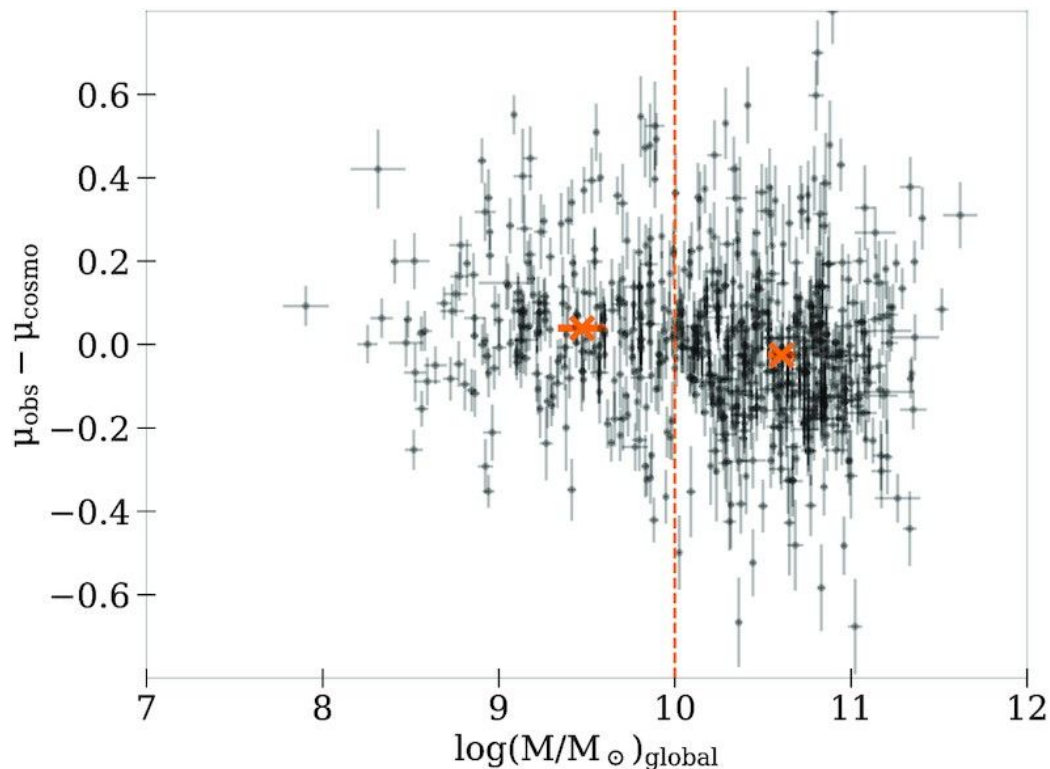
NASA/JPL-Caltech/ESO/R. Hurt
Nicholas B. Suntzeff

$$\mu^s = m_B^s - M + \alpha x_1^s + \beta c^s$$

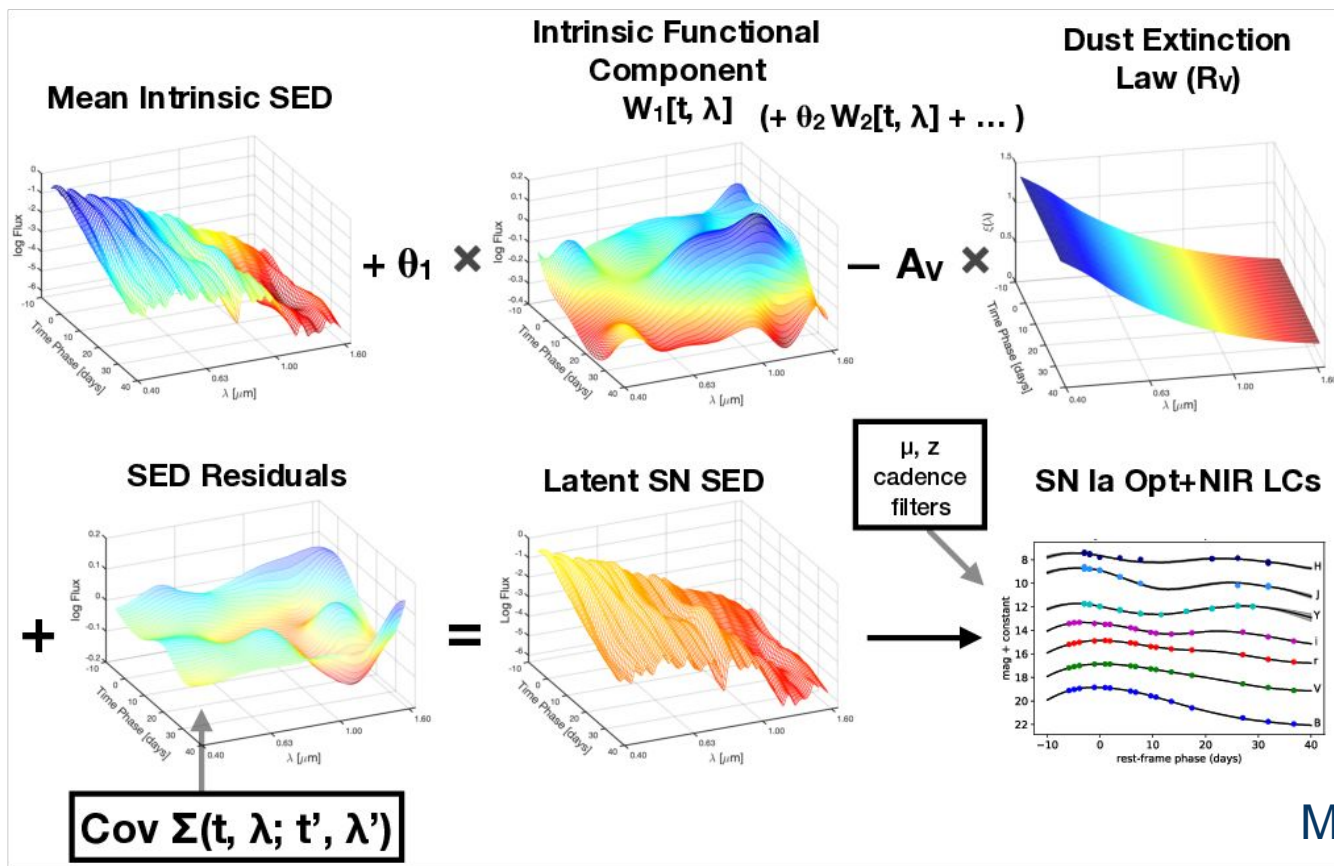
- Tripp formula - one parameter for two separate effects
- Correctly handling dust is key for both SN Ia astrophysics and cosmology
- If intrinsic effect misattributed to dust, could lead to bias

The 'Mass Step'

- Disentangling intrinsic variation from dust vital to understanding host galaxy 'mass step'
- Just different dust properties?
 - SN populations identical, just appear different?
- Or intrinsic differences?
 - Interesting physical implications, different progenitor channels?

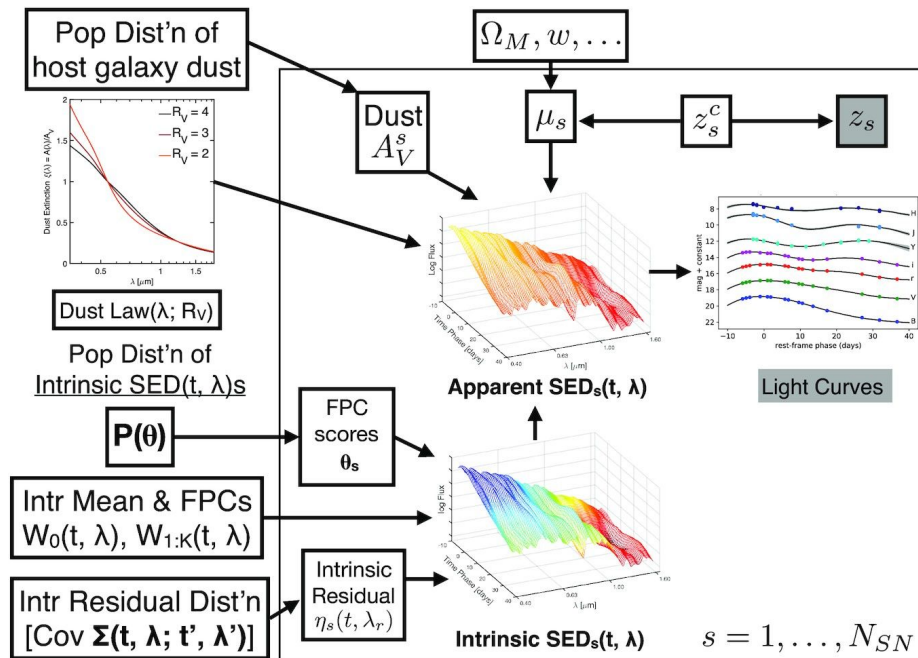


The BayeSN Model



Advantages of BayeSN

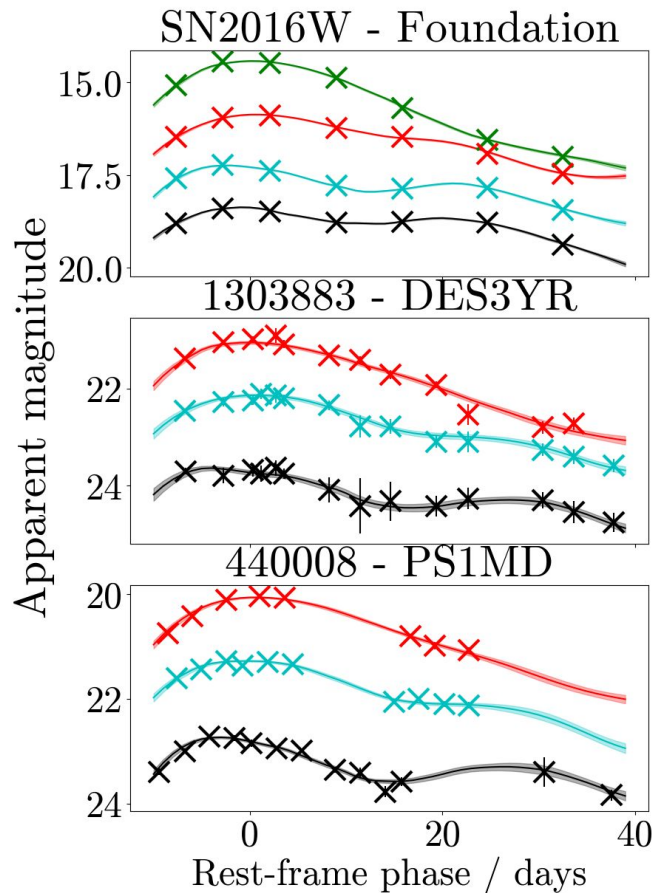
- Hierarchical Bayesian Model
 - Allows for separate treatment of dust and intrinsic variation
 - Can treat SNe on either side of the mass step as separate populations
- Full SED model
 - Uses full light curve in all bands rather than just single colour/magnitude pairs
- Model extends into NIR wavelengths
 - Better SN Ia standardisation and constraint of host galaxy dust



Mandel+2020

Investigating the Mass Step

- Infer properties of SN populations separately on each side of the mass step using BayeSN
- Combined sample of 475 SNe Ia from Foundation, DES3YR and PS1MD
- Jointly infer dust and intrinsic differences, see which is more consistent with data

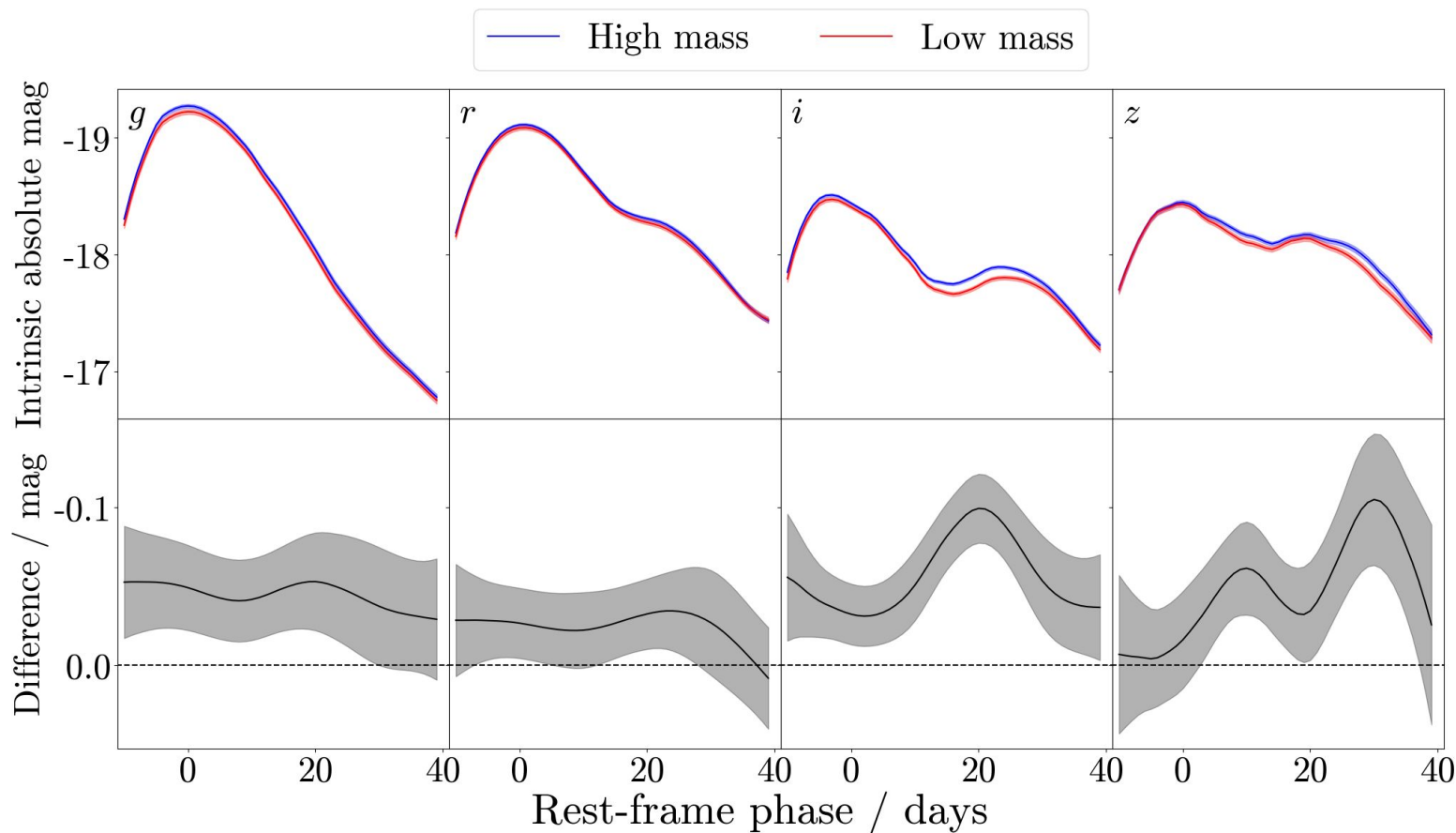


Example
BayeSN
fits

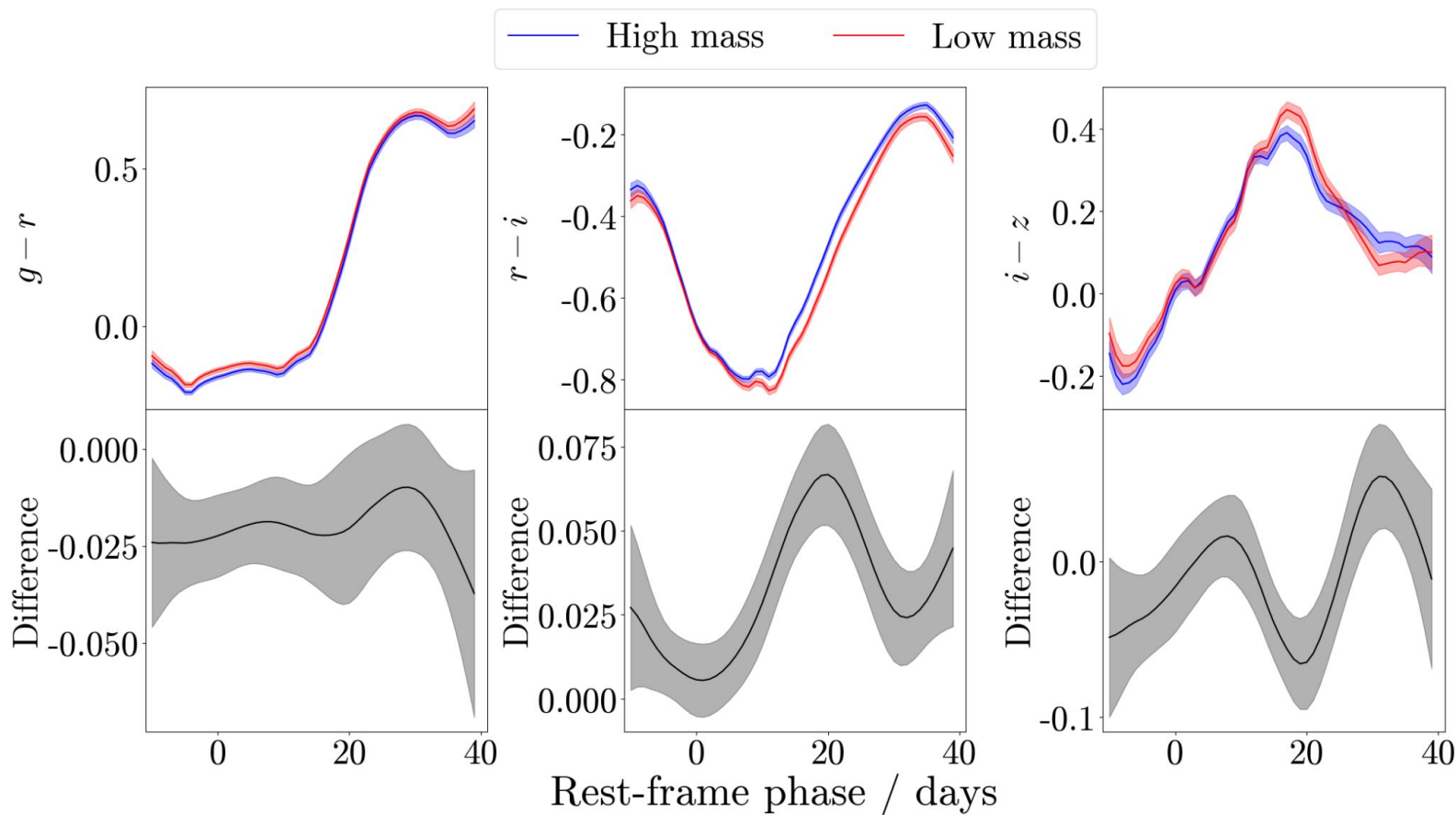
Method	ΔM_0 / mag	$\Delta(g - r)_{\text{peak}}^{\text{int}}$	$\mathbb{E}[R_V]$	
			HM	LM
Intrinsic mag diff.	-0.049 ± 0.016	—	2.51 ± 0.16	2.74 ± 0.35
Intrinsic SED diff.	-0.049 ± 0.027	-0.022 ± 0.010	2.26 ± 0.14	3.36 ± 0.51

- Our results support the existence of intrinsic differences between SNe Ia in each mass bin
- Depending on our assumptions, our results also indicate differences in host galaxy dust properties

Intrinsic Baseline Light Curves



Intrinsic Baseline Colour Curves



Method	ΔM_0 / mag	$\Delta(g - r)_{\text{peak}}^{\text{int}}$	$\mathbb{E}[R_V]$	
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- Our results support the existence of intrinsic differences between SNe Ia in each mass bin
- Depending on our assumptions, our results also indicate differences in host galaxy dust properties
- Lots of different, SED-level effects. Analyses of larger samples will be able to further constrain the relative balance of intrinsic and extrinsic effects

BayeSN Out Now!

New GPU-accelerated, Python implementation of BayeSN out now!

[arXiv:2401.08755](https://arxiv.org/abs/2401.08755), github.com/bayesn/bayesn, pip installable, SNANA integration in progress

 README  MIT license

BayeSN

Hierarchical Bayesian SED modelling of type Ia supernova light curves

Developed and maintained by: Matt Grayling (@mattgrayling), Stephen Thorp (@stevet40), Gautham Narayan (@gnarayan), and Kaisey S. Mandel (@CambridgeAstroStat) on behalf of the BayeSN Team (@bayesn).

About

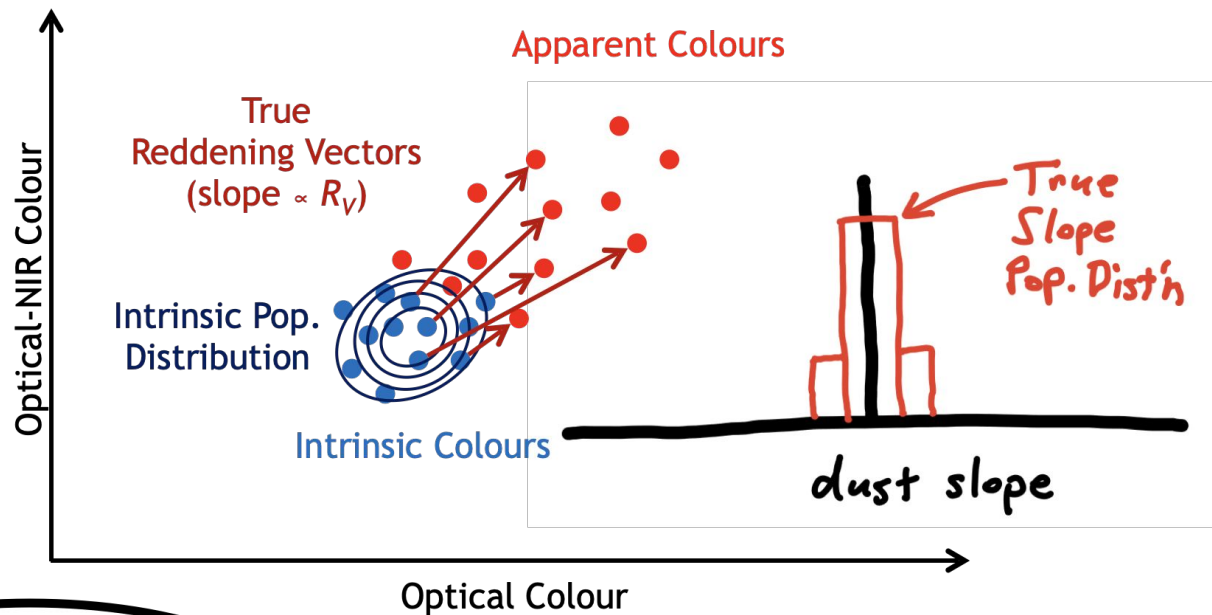
BayeSN is a probabilistic optical-NIR SED model for type Ia supernovae, allowing for hierarchical analysis of the population distribution of physical properties as well as cosmology-independent distance estimation for individual SNe. This repository contains an implementation of the BayeSN SED model built with numpyro and jax, with support for GPU acceleration, as discussed in Grayling+2024 (submitted to MNRAS).



Scan for arXiv link

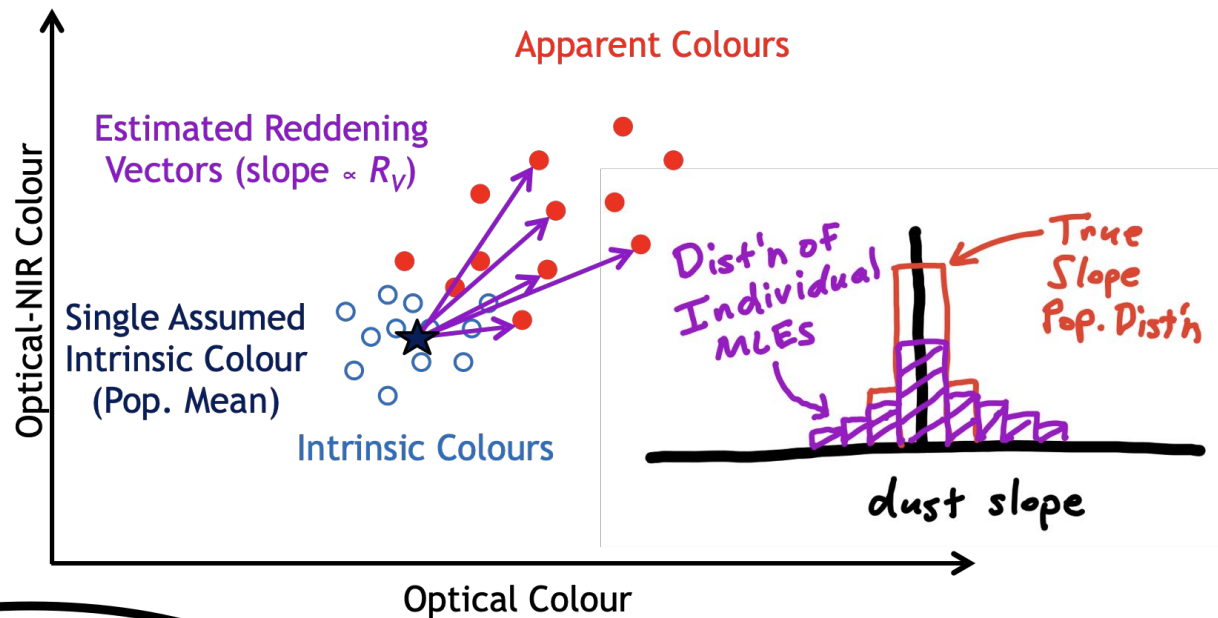
- Using BayeSN, we can use the full time and wavelength evolution of SN Ia light curves to separately constrain intrinsic and extrinsic differences between mass bins
- We find evidence in favour of intrinsic differences, supporting the idea of different SN Ia populations in different environments
- Extrinsic differences may also play a role. Future analyses with larger samples will be able to better constrain the relative size of these effects
- BayeSN out now, pip installable, can also be used to infer distances for cosmological analyses

Why include intrinsic variation?



Need to account for
intrinsic colour scatter
when inferring R_V !

Why include intrinsic variation?



Otherwise width of R_V
distribution will be
overestimated!

- For typical Bayesian inference, each supernova would be fit separately

$$R_V \sim U(1, 5)$$

- For a hierarchical model, we jointly sample the posteriors of global and individual supernova properties

$$R_V \sim N(\mu_R, \sigma_R^2)$$

- We can model intrinsic variation and dust properties at the population level as two separate effects